Radio Frequency Interference Detection and Localization in Sentinel-1 Images

Xiangguang Leng, *Member, IEEE*, Kefeng Ji, *Member, IEEE*, and Gangyao Kuang, *Senior Member, IEEE*

***Abstract*— The C-band Sentinel-1 synthetic aperture radar (SAR) images are affected by radio frequency interference (RFI), and this article proposes an RFI detection and localization method for these images. First, a detailed analysis of RFI based on the generation mechanism is provided, which shows that RFI in the ground range detected (GRD) products differs from common backscattering not only in frequency spectrums but also in radiometric characteristics. Then, an RFI index (RFII) is proposed for RFI detection, which takes full advantage of unique RFI characteristics in dual-polarization GRD images. Finally, both detection results in ascending and descending passes are used to locate the ground RFI sources, assuming that the presence of RFI is persistent. Theoretical analyses show that the location area is a diamond of approximately 88.76 km^2. Experimental results demonstrate that the proposed method performs quite well on Sentinel-1 GRD data. It provides a convenient and tractable tool for assessing Sentinel-1 data quality as well as for monitoring RFI. Thus, Sentinel-1 measurements can be used to monitor C-band RFI, which is an important task of electromagnetic spectrum management.**

***Index Terms*— Interference detection, localization, radio frequency interference (RFI), RFI index (RFII), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) sends out a series of microwave signals of specific frequencies and records the backscattered signals from the Earth's surface. Then, the raw echo data are digitally processed into focused SAR data [1]–[3]. However, the frequencies can also be used by other applications, such as frequency modulation broadcast, television, communication, and missile guidance. These signals may be captured by SAR sensors and, thus, degrade the SAR imaging quality severely [4]. These kinds of signals are called radio frequency interference (RFI). Usually, SAR of low frequencies (especially the very high frequency (VHF)/ultra high frequency (UHF) band) is badly impacted by RFI. L- to C-bands which are often used in spaceborne SAR can be also affected by RFI.

On the one hand, since SAR imaging quality is severely degraded by interference, the suppression of RFI is an important issue. A large number of RFI suppression methods were proposed in the literature, such as methods based on decomposition, filtering, sparse model, low-rank model, and time–frequency analysis [5]–[9]. On the other hand, the detection and localization of RFI is an important task of electromagnetic spectrum management. Several sensors have been used to survey RFI. The CubeRRT mission has already been deployed and finished its operational life for monitoring the whole 6–40-GHz spectrum [10]. SMAP, Aquarius, ALOS, and SMOS data are used to produce L-band RFI maps [11]–[15]. ASCAT scatterometers, ENVISAT, and RadarSAT were partly used to monitor C-band RFI [16], [17].

Sentinel-1, which was launched by the European Space Agency (ESA), is an advanced Earth observation satellite carrying an SAR imaging instrument [18], [19]. It is a dual-polarization system, working in the C-band (5.405 GHz, corresponding to a radar wavelength of about 5.6 cm) with four modes: interferometric wide swath (IW), stripmap (SM), extra wide swath (EW), and wave (WV). Sentinel-1 is composed of a constellation of two satellites, Sentinel-1A and Sentinel-1B. They share the same orbital plane with a 180° orbital phasing difference, making it possible for interferometry and rapid revisit. Due to the high-quality imaging and free distribution policy [20], Sentinel-1 has been widely used in maritime monitoring, land monitoring, emergency management, and mapping applications [21]–[26].

However, since RFI removal in the processing from raw data (Level 0) to single look complex (SLC) data (Level 1) is not performed by the Sentinel-1 Instrument Processing Facility (IPF), Sentinel-1 images are severely affected by C-band RFI [27], [28]. Santamaria *et al.* [22] found that RFI can cause a large number of false alarms in mass processing of Sentinel-1 images for ship detection, and the manual labeling of areas polluted by RFI is quite time-consuming. Thus, part of Sentinel-1 images is discarded. Leng *et al.* [29]–[31] also found that the dark sea surface in Sentinel-1 images is affected by RFI and it can degrade the detection performance. Monti-Guarnieri *et al.* [4] provided an RFI map over Europe based on Sentinel-1 echo data from the aspect of electromagnetic spectrum management.

Usually, the spectrum characteristics of RFI are different from SAR signals. RFI in Sentinel-1 images also presents

Manuscript received July 24, 2020; revised October 7, 2020, November 28, 2020, and December 28, 2020; accepted December 29, 2020. Date of publication January 26, 2021; date of current version October 26, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 62001480 and Grant 61971426. *(Corresponding author: Kefeng Ji.)*

The authors are with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, National University of Defense Technology, Changsha 410073, China, and also with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: luckight@163.com; jikefeng@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3049472

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

different complex-valued statistics [29]–[31]. Thus, most RFI suppression or identification methods are based on spectrum analyses of raw data or SLC data for granted. Rare literature discusses using ground range detected (GRD) products (Level 2) to monitor RFI. However, these methods usually involve in complicated and time-consuming spectrum estimation process. Besides, some applications are preferred to operate on GRD images [32]–[36] since they are more operational, e.g., the massive ship detection presented by Santamaria *et al.* [22]. Moreover, raw data or SLC data cost much more storage space than GRD products. For example, unzipped Sentinel-1 IW SLC data acquire approximately 7.5 GB, while GRD data acquire only approximately 1.5 GB. This has imposed a heavy burden on the common desktop users when dealing with massive data. Thus, an RFI detection and localization method based on GRD images is urgently needed.

This article proposes a new RFI detection and localization method in Sentinel-1 products. It takes full advantage of the different characteristics of RFI in dual-polarization Sentinel-1 GRD images. Experimental results based on Sentinel-1 GRD data demonstrate that the proposed method is quite effective. The main contribution of this article is concluded as follows.

1) We provided a detailed characterization of RFI based on the generation mechanism analyses. It shows that RFI differs from common backscattering not only in spectrums but also in radiometric characteristics of GRD images. The underlying rationale to detect and locate RFI based on GRD images is elaborated.
2) A powerful RFI index (RFII) is first proposed. It is designed based on the unique characteristics of RFI in dual-polarization Sentinel-1 GRD images. It can be used to indicate RFI strength and detect strong RFI signals.
3) A complete and operational RFI detection and localization workflow in Sentinel-1 images is presented. It can be used to detect and locate ground RFI sources into a diamond of approximately 88.76 km^2.
4) An example is given and discussed. It demonstrates that the proposed method can provide a convenient and tractable measure for assessing Sentinel-1 data quality as well as for monitoring RFI.

The remainder of this article is organized as follows. Section II presents the characterization of RFI based on the generation mechanism and proposes the detection and localization method. Section III gives an introduction to the experimental data. Section IV presents the experimental results and discusses the further applications of the proposed method. Finally, Section V concludes this article.

II. Model and Methodology

In this section, the generation mechanism of RFI in Sentinel-1 images is introduced first. Then, the characteristics of RFI and the rationale of detection and localization are elaborated. Finally, the RFI detection and localization workflow is proposed based on the unique RFI characteristics.

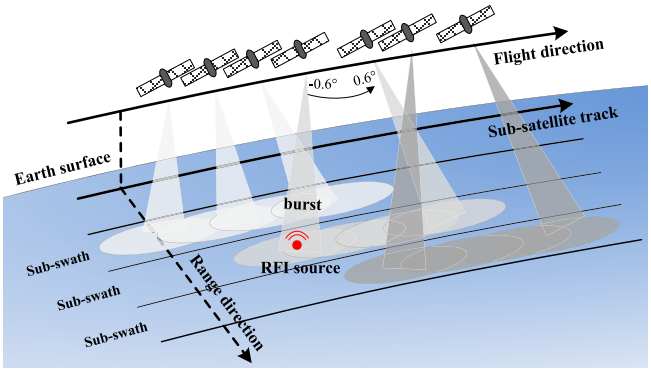


Fig. 1. Illustration of Sentinel-1 TOPSAR IW imaging mode.

A. Generation Mechanism of RFI in Sentinel-1 Images

The generation mechanism of RFI is quite different from common backscattering from the Earth. RFI signals are usually expected to be a linear combination of multiple single frequencies caused by radiation sources. They are usually considered to be narrowband with strong power. RFI signals can be modeled as [37]

$$\mathrm{rfi}(\tau) = \sum_{l=1}^{L} A_l(\tau) \exp[j(2\pi f_l \tau + \theta_l)] \tag{1}$$

where A_l, f_l, and θ_l are the amplitude, carrier frequency, and initial phase of the lth interference component, respectively. τ is the fast time. L is the number of interference components. RFI signals are independent of SAR echoes. Therefore, an SAR pulse contaminated by RFI at each distance line can be modeled as

$$x(\tau) = s(\tau) + \mathrm{rfi}(\tau) + n(\tau) \tag{2}$$

where $s(\tau)$ is the SAR echo and $n(\tau)$ is the additive noise.

The IW mode is the main operation mode for Sentinel-1. Since RFI mainly appears in Sentinel-1 IW mode images, we address Sentinel-1 IW mode here. The IW mode can provide large swaths (approximately 250 km) with large incidence angle ranges. As shown in Fig. 1, it benefits from the Terrain Observation with Progressive Scans SAR (TOPSAR) technique, which can capture three subswaths at once. Each subswath image consists of a series of bursts. At the beginning of each burst, as shown in Fig. 1, the antenna beam is pointed backward of $-0.6°$ and sweeps forward up to $+0.6°$ at the burst end. The system starts acquiring since the transmission of the very first pulse [4]. During the acquisition, if there is an RFI source in the area covered by a burst, as shown in Fig. 1, the SAR imaging could be affected by RFI. This is also the reason why RFI usually only occurs in specific bursts of a subswath.

B. Characterization of RFI in Sentinel-1 Images

SAR uses 2-D matching-filtering to form high-resolution images. This has the inherent ability in interference suppression. However, strong RFI could still blur useful signals and appear bright stripes along the range direction in SAR images, as shown in Fig. 2(a). Some of them are brighter than land and even man-made objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

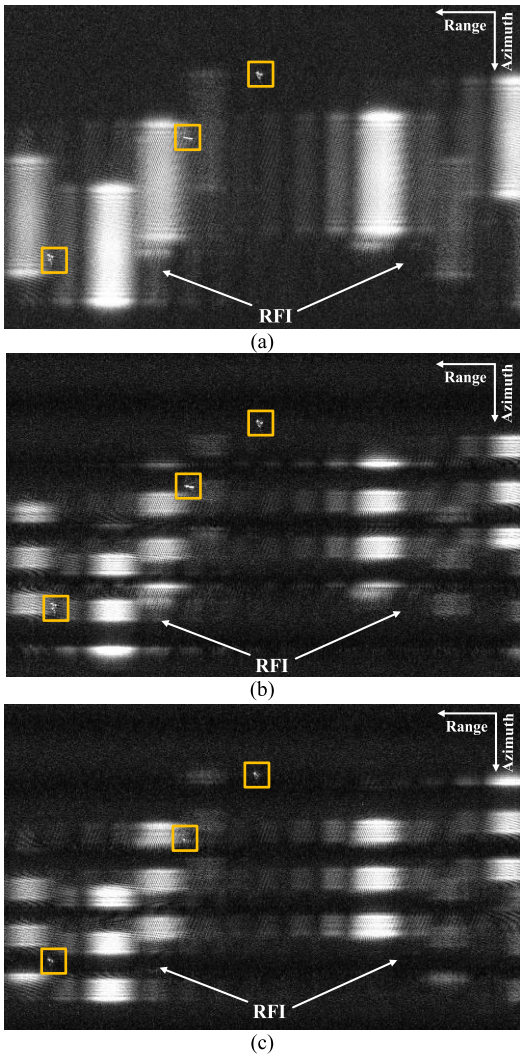


Fig. 2. RFI in Sentinel-1 SLC and sublook images. (a) Original image. (b) and (c) Sublook imges. The bright features are RFI. The yellow rectangles represent ships.

In the Sentinel-1 raw data, RFI occurs only in a few range lines. There are approximately 1700 range lines per second (i.e., the pulse repetition frequency). The increase in energy caused by RFI within each range line is not continuous. It occurs at specific times/ranges and thus gives the pattern along the range direction [27], [28]. Since this pattern depends on the ground RFI source, the intensity, shapes, and textures can be different in different images. RFI in Sentinel-1 images can be divided into several segmentations by textures and like bright "fluorescent lamp" or "zebra crossing" along the azimuth direction. Besides, RFI usually appears much stronger in cross-polarization than co-polarization due to the lower radar cross section of cross-polarization images.

Meanwhile, we find that the same RFI can be received by Sentinel-1 repeatedly during the acquisition. If we separate the image along the azimuth direction into two sublook images, we would see the similarities and differences between the sublook images and the original image further. As shown in Fig. 2, it is interesting that the RFI patterns in the original image are separated into two sublook images. The textures of RFI in sublook images are the same but the locations interlace along with the azimuth direction.

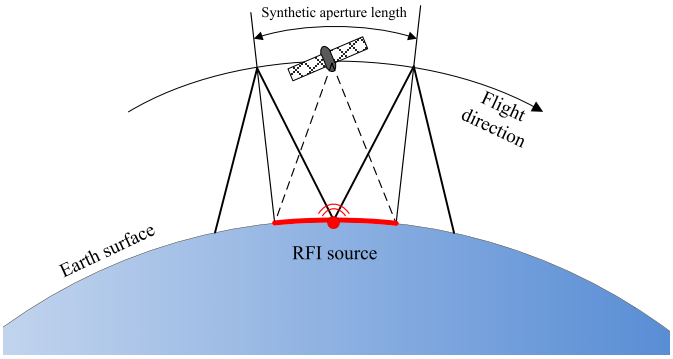


Fig. 3. RFI dispersion along with the synthetic aperture.

TABLE I
SUMMARY OF RFI RADIOMETRIC CHARACTERISTICS IN SENTINEL-1 IMAGES

Image characteristics	Content
Pattern	(1)The shapes and textures can be different depending on the ground RFI source. (2)They can occur repeatedly along the range direction and can be separated along the azimuth direction.
Intensity	(1)The power can be very strong. Some of them are brighter than land and even man-made objects. (2)They usually appear much stronger in cross-polarization than co-polarization.
Size	(1)The length along the range direction could be as large as the coverage of the sub-swath. (2)The length along the azimuth direction would be similar to the synthetic aperture length.

As for the size of RFI in Sentinel-1 images, since it can occur repeatedly along the range direction in one of the sub-swaths, the length along the range direction could be as large as the coverage of the subswath. The azimuth compression part of the SAR imaging takes the received energy over the synthetic aperture (approximately 5 km in azimuth) to form a single pixel, as shown in Fig. 3. Thus, if there is a large increase in energy caused by RFI for any of the range lines, the pixels affected by the energy increase in the azimuth would appear bright. Thus, the length of RFI along the azimuth direction would be similar to the synthetic aperture length [28], as shown in Fig. 3.

In sum, despite the difference in spectrums, the RFI in Sentinel-1 images has unique radiometric characteristics in terms of the pattern, intensity, and size. Table I gives the summary.

C. Proposed Detection and Localization Method

Based on the unique radiometric characteristics of the RFI in Sentinel-1 GRD images, we propose an RFII for detection and a localization methodology in this section. Then, a complete RFI detection and localization method is provided.

1) RFI Index: Since Sentinel-1 is a dual-polarization system, a beneficial radiometric characteristic that can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

used to detect RFI is the difference between the cross- and co-polarization channels. The detection rationale is that most common backscattering from the Earth appears much stronger in co-polarization than cross-polarization, while this is not true for RFI. Thus, this difference can be used to detect RFI. However, it should be noted that strong depolarization areas of the vegetation or sea [38] may cause false alarms. They can be excluded since they usually appear darker than RFI in the cross-polarization polarization channel.

Taking into consideration of these points, an RFII is proposed

$$\mathrm{RFII} = R\left(\frac{\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}}{\sigma^0_{\mathrm{Co}}}\right) H\left(\sigma^0_{\mathrm{Cross}} - \mathrm{Tol}\right) \tag{3}$$

where $\sigma^0_{\mathrm{Cross}}$ and σ^0_{Co} represent calibrated backscatter coefficients of the cross- and co-polarization channels, respectively. Tol is the tolerance coefficient of the potential RFI to exclude the dark vegetation and sea areas. $R(\cdot)$ is the unit ramp function and $H(\cdot)$ is Heaviside step function, or the unit step function

$$R(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{4}$$

$$H(x) = \frac{d}{dx} R(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0. \end{cases} \tag{5}$$

$H(0) = 0$ is defined as a complement here. The effect of $R(\cdot)$ here is to preserve the positive values of the variable. The effect of $H(\cdot)$ is to reject negative values.

In summary, the first term of the RFII, i.e., $R\left(\left(\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}\right)/\sigma^0_{\mathrm{Co}}\right)$, is used to prescreen potential RFI pixels in a fast way. It can be used to detect RFI alone but may cause false alarms. The second term $H\left(\sigma^0_{\mathrm{Cross}} - \mathrm{Tol}\right)$ is used to discard the strong depolarization scattering effect caused by the vegetation or sea areas.

Since $R(x) = xH(x)$, the RFII can also be written as

$$\mathrm{RFII} = \frac{\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}}{\sigma^0_{\mathrm{Co}}} H\left(\frac{\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}}{\sigma^0_{\mathrm{Co}}}\right) H\left(\sigma^0_{\mathrm{Cross}} - \mathrm{Tol}\right). \tag{6}$$

Since $H\left(\frac{\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}}{\sigma^0_{\mathrm{Co}}}\right) = H\left(\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}\right)$, it can also be rewritten as

$$\mathrm{RFII} = \frac{\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}}{\sigma^0_{\mathrm{Co}}} H\left(\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}\right) H\left(\sigma^0_{\mathrm{Cross}} - \mathrm{Tol}\right). \tag{7}$$

This formula variation can also help understand the RFII in three terms. The first term $\left(\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}\right)/\sigma^0_{\mathrm{Co}}$ is the main coefficient. The second and third terms, $H\left(\sigma^0_{\mathrm{Cross}} - \sigma^0_{\mathrm{Co}}\right)$ and $H\left(\sigma^0_{\mathrm{Cross}} - \mathrm{Tol}\right)$, are used to reject common backscattering areas from the Earth and the strong depolarization areas of the vegetation or sea, respectively. The larger the tolerance coefficient, the larger number of potential RFI results are discarded.

2) Localization Method: The Sentinel-1 has a repeat cycle of 12 days by one satellite. Taking into consideration both passes (ascending and descending) and the overlap, two satellites can revisit an area in 1–3 days [19] (three days at the equator and less than 1 day at high latitudes). Thus, Sentinel-1 can monitor RFI continuously. Assuming that the presence of RFI is persistent in ascending–descending interval, the unique image characteristics of RFI in Sentinel-1 images can be utilized to locate the ground RFI source. The rationale is described as follows.

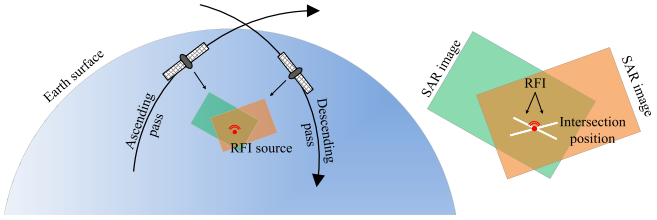


Fig. 4. Illustration of the RFI localization method.

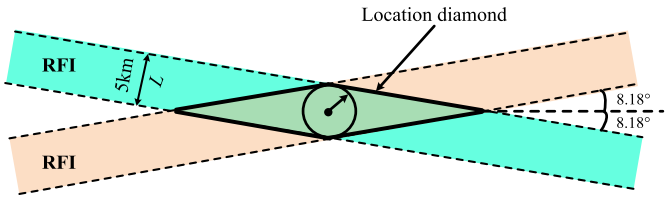


Fig. 5. Illustration of the RFI localization precision.

As mentioned above, though RFI occurs only in a few range lines in the raw data, the azimuth extent of the RFI in focused SAR images shows the synthetic aperture length, i.e., the orientation of RFI is determined by the azimuth direction or the flight direction. One single image cannot give a precise position since it is ambiguous in the azimuth direction. However, if the same RFI ground source is recorded by two images of different passes, the ambiguity can be eliminated by the intersection of two images with different azimuth directions. As shown in Fig. 4, the crosspoint of the two RFI can be considered as the location of the ground RFI sources. Thanks to the TOPSAR mode, the ground footprint is short, and this ensures a good localization.

Since the orbital inclination α of Sentinel-1 is 98.18°, the level inclination of the RFI between the range direction is approximately 8.18° (i.e., $\alpha - 90°$), as shown in Fig. 5. Each Sentinel-1 satellite is in a near-polar, sun-synchronous orbit and Sentinel-1 IW images mainly cover the low- and mid-latitudes. Thus, the cross-angles between ascending and descending beams for different latitudes can be considered as approximately the same. If we draw two dashed lines as the contours for each RFI, the intersection of the RFI can be identified in the diamond, as shown in Fig. 5. The area of this diamond is

$$A \approx L^2\left(\tan(\alpha - 90°) + 1/\tan 2\alpha\right) \approx 88.76\ \mathrm{km}^2 \tag{8}$$

where L is the synthetic aperture length and α is the orbital inclination. Thus, the location area is a diamond of approximately 88.76 km^2.

However, it should be noted that Sentinel-1 can also be affected by other C-band satellites, e.g., the RadarSAT-2 or GF-3 satellites, though these cases are much rarer than the ground sources. The RFI caused by these satellites cannot be located by this method since they are not the ground sources.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
EXPERIMENTAL DATA DESCRIPTION

Image Name	Image ID	Polarization	Pass	Size	Sensing time
S0	S1A_IW_GRDH_1SDV_20190703T141637_20190703T141702_027954_032807_0268	VV&VH	Ascending	25833×16724	2019/07/03 14:16 UTC
S1	S1A_IW_GRDH_1SDV_20190708T142449_20190708T142514_028027_032A4E_326D	VV&VH	Ascending	25817×16728	2019/07/08 14:24 UTC
S2	S1A_IW_GRDH_1SDV_20190711T021528_20190711T021552_028063_032B59_2FBB	VV&VH	Descending	25723×15607	2019/07/11 02:15 UTC
S3	S1A_IW_GRDH_1SDV_20190715T141638_20190715T141703_028129_032D53_3841	VV&VH	Ascending	25823×16725	2019/07/15 14:16 UTC

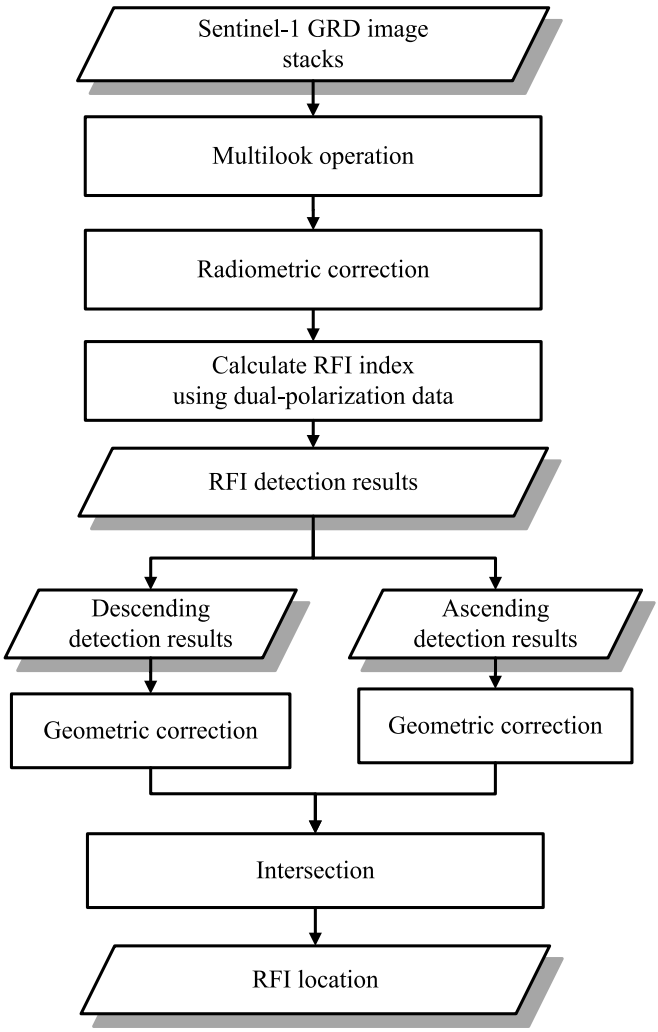


Fig. 6. Flowchart of the proposed method.

3) Proposed Detection and Localization Flow: Based on the detection and localization rationales, a brief illustration of the proposed detection and localization work flow is presented in Fig. 6, which can be concluded as follows.

1) *Input Sentinel-1 GRD Image Stacks*:
To detect and locate the potential RFI in an area, the Sentinel-1 GRD image stacks should cover the same area and include both acquisitions of ascending and descending passes.
2) *Multilook Operator*:
Generally, an SAR image appears inherent speckle noise. Since the images are GRD products, we implement the space-domain multilook method by averaging a single look image with a sliding window. This may reduce false alarms caused by the speckle noise. The number of range and azimuth looks, i.e., the size of the sliding window, can be set as approximately 20.
3) *Radiometric Correction*:
The Sentinel-1 GRD data do not include radiometric corrections and significant radiometric bias remains. Therefore, it is necessary to apply the radiometric correction to SAR images so that the pixel values represent the true radar backscatter. Besides, the comparison of multitemporal Sentinel-1 images also needs the radiometric correction. σ^0 calibration is applied here.
4) *Calculate RFII Using Dual-Polarization Data*:
The RFII is calculated based on (3). The tolerance coefficient Tol here is recommended to be 0.005 empirically. This value is based on the fact that most calibrated backscatter coefficients of the forest and the sea clutter in cross-polarization images are smaller than 0.005. Thus, false alarms caused by them can be eliminated. This is a tradeoff setting to detect strong RFI and discriminate false alarms.
5) *Geometric Correction of Descending and Ascending Detection Results*:
After RFI is detected, both descending and ascending detection results are geometrically corrected. This can transform the vertical or horizontal RFI in Sentinel-1 images into different directions.
6) *Intersection for RFI Localization*:
Finally, the detection results after geometric correction can be intersected. The ground RFI sources are considered to be located in the intersection part.

III. EXPERIMENTAL DATA DESCRIPTION

To validate the performance of the proposed method, a meaningful and challenging data set is tested. The data set consists of four GRD products acquired from Sentinel-1, which are named S0, S1, S2, and S3, respectively. Detailed information of the data set is listed in Table II. The pseudo-color images are presented in Fig. 7. The geolocations of these products are shown in Fig. 8.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

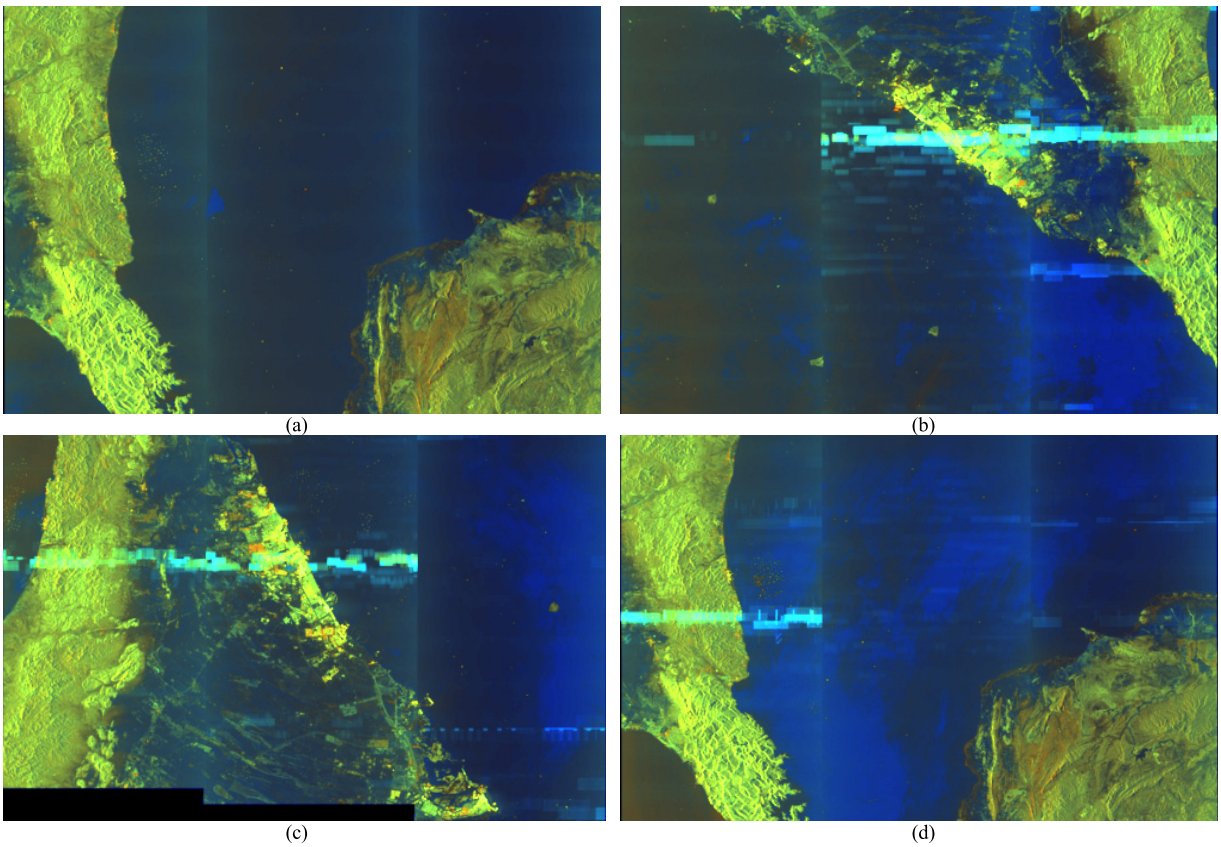


Fig. 7. Pseudo-color images acquired over the Strait of Hormuz and the Gulf of Oman. (a)–(d) Labeled as S0, S1, S2, and S3, respectively, in Table II.

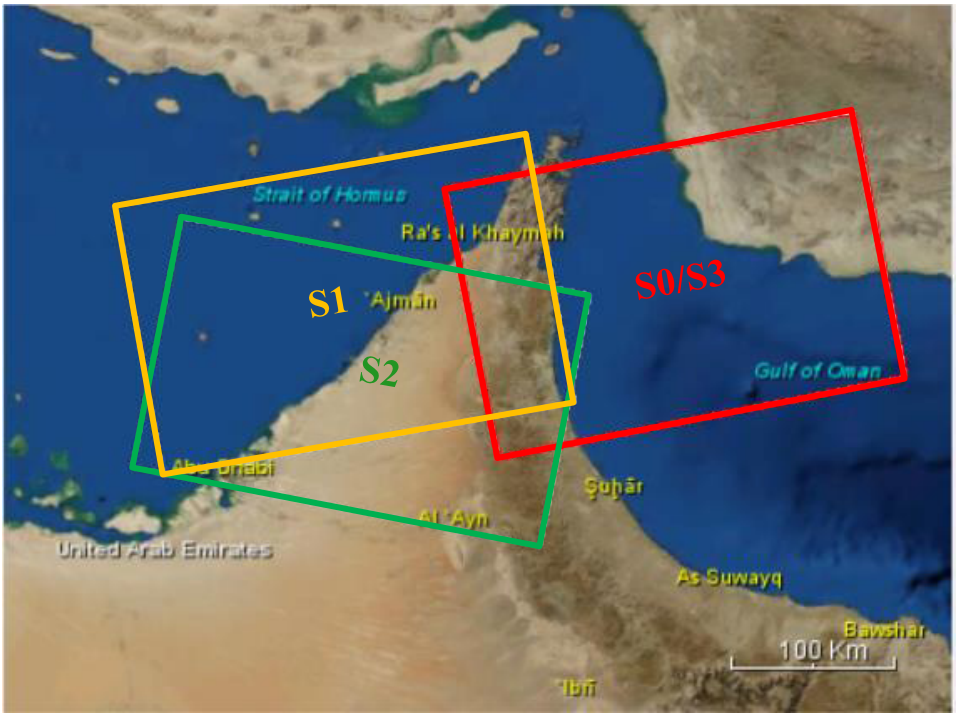


Fig. 8. Geolocations of the experimental data. S0 (S3), S1, and S2 are represented in red, yellow, and green, respectively.

It can be seen that these images were acquired by Sentinel-1 A and have a dual-polarization of VV and VH. They covered a wide area over the Strait of Hormuz and the Gulf of Oman. They contain complex scenes such as urban, sea, vegetation, and bare areas.

S0 and S3 have the exactly same footprint. Based on the eye interpretation, S0 can be regarded as a reference for RFI-free Sentinel-1 images. The change between S0 and S3 can clearly show the occurrence of RFI. S2 was acquired in a descending pass, while S0, S1, and S3 were acquired in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

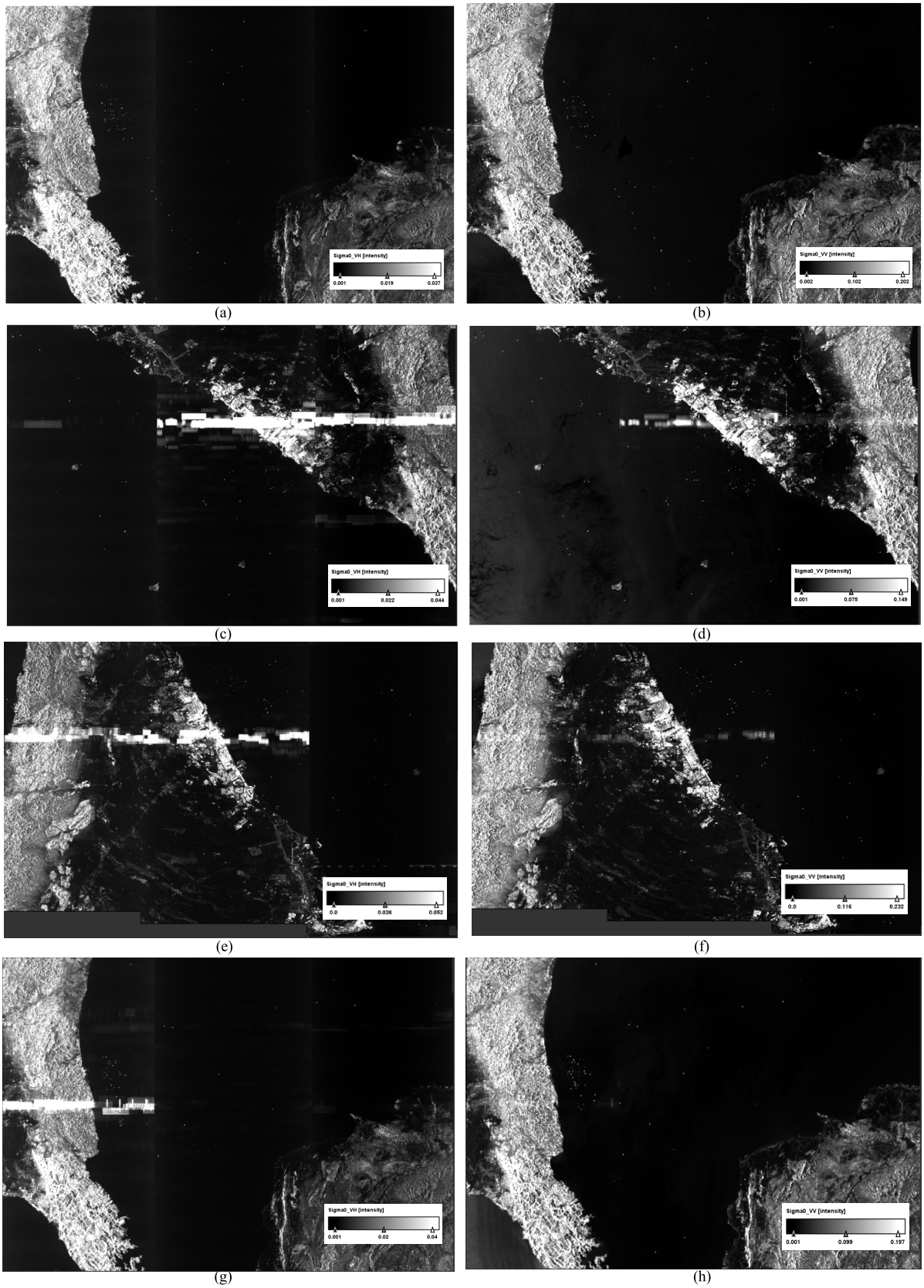


Fig. 9. Experimental data after radiometric calibration. (a) S0 VH. (b) S0 VV. (c) S1 VH. (d) S1 VV. (e) S2 VH. (f) S2 VV. (g) S3 VH. (h) S3 VV. RFI in VH mode images is much stronger than VV mode images. Some RFI in VV mode images can be invisible.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

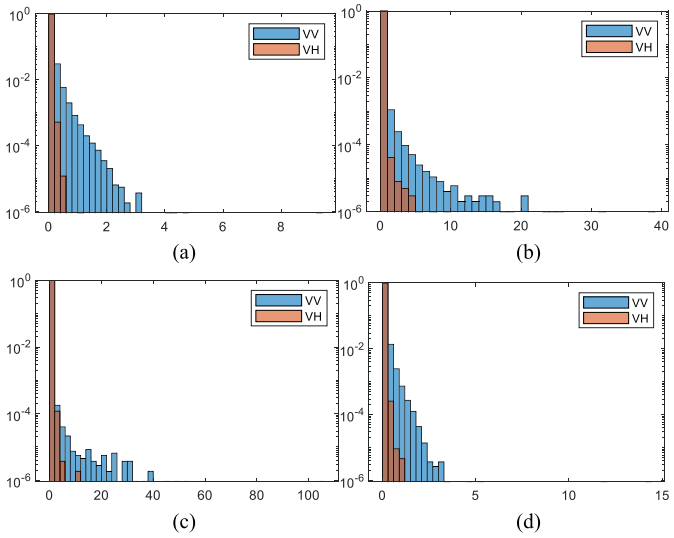


Fig. 10. Histograms of the calibration results. (a) S0. (b) S1. (c) S2. (d) S3. The main parts of VV images are much stronger than VH images.

ascending passes. These images were sensed from July 3, 2019 UTC to July 15, 2019 UTC. The shortest interval is three days. The image sizes are approximately 25 000 × 16 000.

IV. Experimental Results and Discussion

In this section, the experimental results acquired by the proposed method are introduced and analyzed. A further discussion of the proposed method is also provided.

A. Experimental Results and Analysis

Fig. 9 shows the dual-polarization radiometric calibration results of S0, S1, S2, and S3. Images in the first column are the VH results and the second are the VV results. Fig. 10 shows the histograms of the calibration results. It can be seen that as expected, the main parts of VV images are much stronger than VH images. Most VV values are five times larger than VH values. However, it can be seen from S1, S2, and S3 that RFI in VH mode images is much stronger than VV mode images. Some RFI in VV mode images can be invisible.

Fig. 11 shows the RFI detection results of the experimental data. The RFI strength in RFII maps of S1, S2, and S3 is presented in different colors. It can be seen that though there are complex scenes in these images, the proposed RFII can indicate RFI correctly. The strong RFI areas are all effectively detected. Specifically, S0 is free of RFI. S1, S2, and S3 are all contaminated by RFI. The second and third subswaths of S1 are affected by strong RFI. At least four bursts of the third subswath have presented RFI. All three subswaths of S2 are affected by RFI. RFI in the first and second subswath may be affected by the same ground RFI source since they are in the same azimuth lines. The first subswath of S4 is strongly affected by RFI. The third subswath also present RFI. Since S3 has the same footprint of S0, it can be known that this RFI occurs during July 3, 2019 and July 8, 2019 UTC.

It can be also known that no false alarms caused by the strong depolarization areas of the vegetation or sea areas are wrongly detected. This has the benefit of the tailored second term of RFII in (3). If the second term of RFII in (3) is removed, the RFII maps could contain a large number of false alarms. For example, Fig. 12 shows the RFII results on image S3 without including this second term. It can be seen that, although a larger number of RFI occurrences are identified, the extra detected potential RFI is very weak, with values of RFII smaller than 0.2 and very likely to contain false alarms, as those isolated points shown in the yellow rectangle caused by the sea. The probability of false alarms (PFA) is approximately 2.19% for the result in Fig. 12. PFA here is defined as $P_{\mathrm{f}} = N_{\mathrm{fd}}/N_{\mathrm{bg}}$, where N_{fd} denotes the number of false detected backscatters and N_{bg} denotes the number of all background backscatters excluding true RFI backscatters. The presence of these false alarms demonstrates that the second term is essential in the design of the RFII. It should be noted that here the tolerance coefficient Tol in the second term is set to 0.005 empirically; however, the detection result can be adjusted by changing the value of Tol.

All detection results are geometrically corrected and then are overlaid on Google Earth, as shown in Fig. 13. We use white solid lines to mark the detected RFI. It can be seen that RFI has affected a wide range of areas. Table III has shown the percentages of images affected by RFI. It can be seen that the area of S1 affected by RFI is the largest. As mentioned above, S2 is the only image acquired in the descending pass. Thus, the RFI in S2 has the opposite direction to S1 and S3. The level inclination of the RFI between the range direction is approximately 8.18°, as shown in Fig. 12. The RFI in S1 and S3 has the same direction. Some of them could be caused by the same ground source since they are in the same line, as shown in Fig. 13.

The main intersection determined by S2 and S1 or S3 is marked in the white rectangle. The area in the white rectangle is zoomed and more clearly presented in Fig. 14. The intersection area of RFI can be clearly identified in the diamond, as shown in Fig. 14. The center is approximately at 25°11′ 43.59′ N, 55°25′ 43.87″ E. This location is very near the center of Dubai. The measured radius is 2.5 km approximately, which is in accordance with the theoretical analysis.

B. Discussion

For a long time, the conventional RFI detection and localization in SAR images are based on Level 0 or Level 1 data since they rely on the spectrum analysis. This article has presented an RFI detection and localization method which can be operated on the Level 2 data, i.e., the Sentinel-1 GRD products. This merit makes the proposed method a quite tractable and operational method in real applications.

Despite the simple RFI detection and localization in a local area, the proposed method motivates us to do more meaningful tasks as follows.

1) A global C-band RFI map can be produced and updated over time. As the main mode of operations, Sentinel-1 IW images cover the low- and mid-latitudes (up to 60°). This ensures that RFI of most important cities can be recorded. Besides, the occurrence or disappearance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

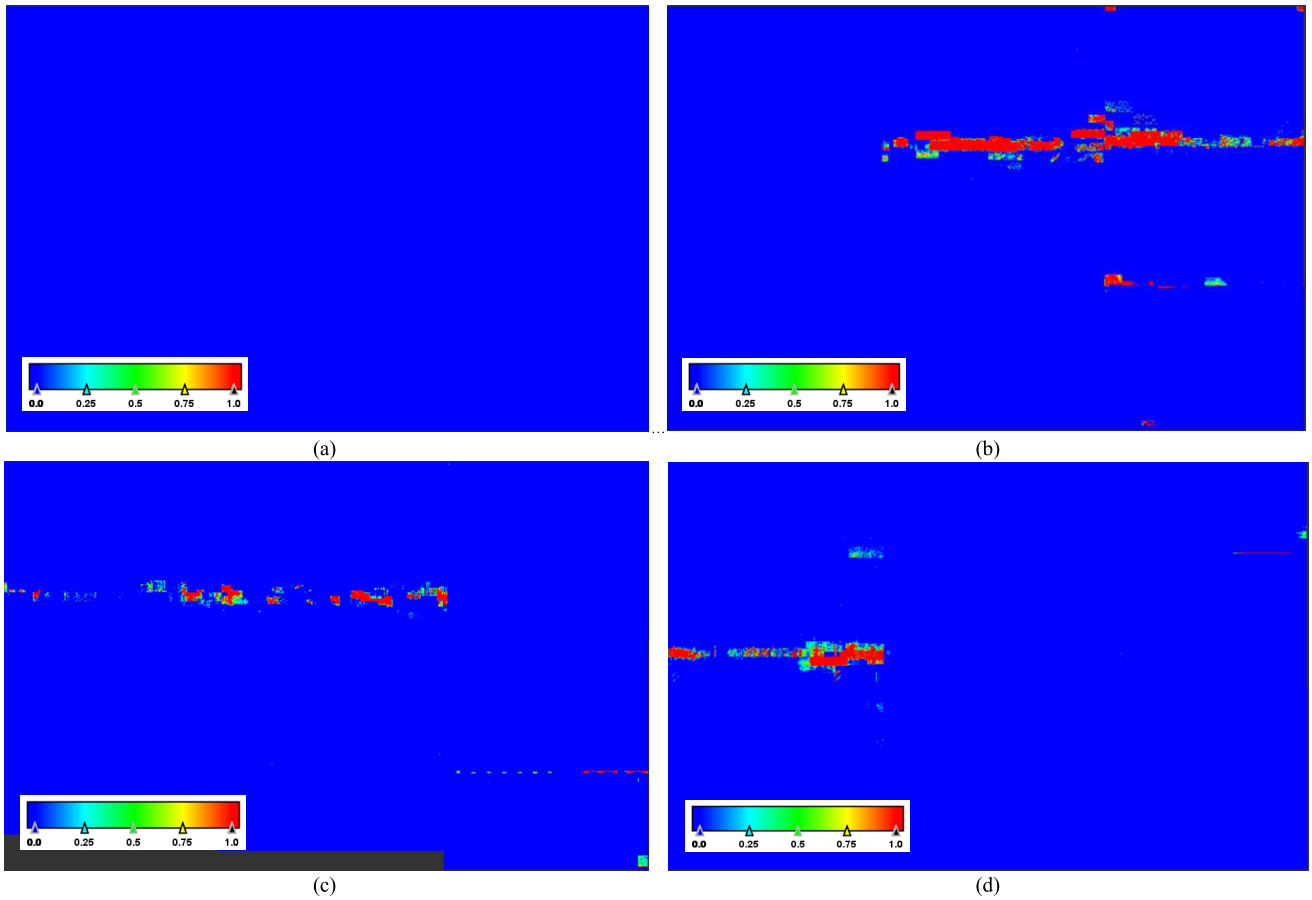


Fig. 11. RFI detection results of experimental data. (a) S0. (b) S1. (c) S2. (d) S3.

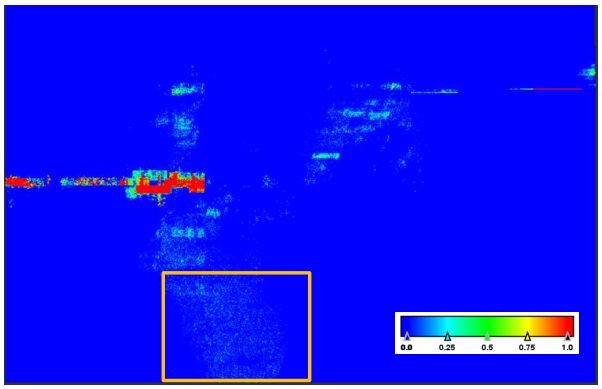


Fig. 12. RFI detection results of S3 without the second term of (3). It contains a large number of isolated false alarms caused by the sea in the yellow rectangle.

interference in an area may imply a potentially important issue. This is quite important for policymakers.

2) It can be used to assess the quality of SAR images. This is essential for some applications, e.g., ship detection or wind field retrieve. The areas strongly contaminated by RFI can be figured out and discarded automatically without a time-consuming manual inspection. Thus, it would reduce or eliminate false alarms or wrong results caused by RFI. It is also data-economic and can avoid data wasting.

TABLE III
PERCENTAGE OF IMAGES STRONGLY AFFECTED BY RFI

Image Name	Percentage affected by RFI
S0	0.00%
S1	2.26%
S2	0.95%
S3	1.12%

3) It can provide new insight into the RFI localization. Though SAR is not specifically designed for RFI monitoring, it can be used to locate the ground RFI source. From the RFI localization model (8), it can be seen that if the synthetic aperture length is small enough and the orbital inclination reaches 135°, the localization precision can be very high.

However, since the proposed RFII is built on a kind of statistics, i.e., RFI is usually stronger in the VH channel than VV channel, it may not work for some particular situations. Though this case is very rare, it may exist that RFI in the VV channel is stronger. Besides, when RFI overlay urban areas of strong reflection, the mixed signals are determined by the scattering of the urban areas, and thus RFI may be not detected. This fact implies that some urban areas would

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

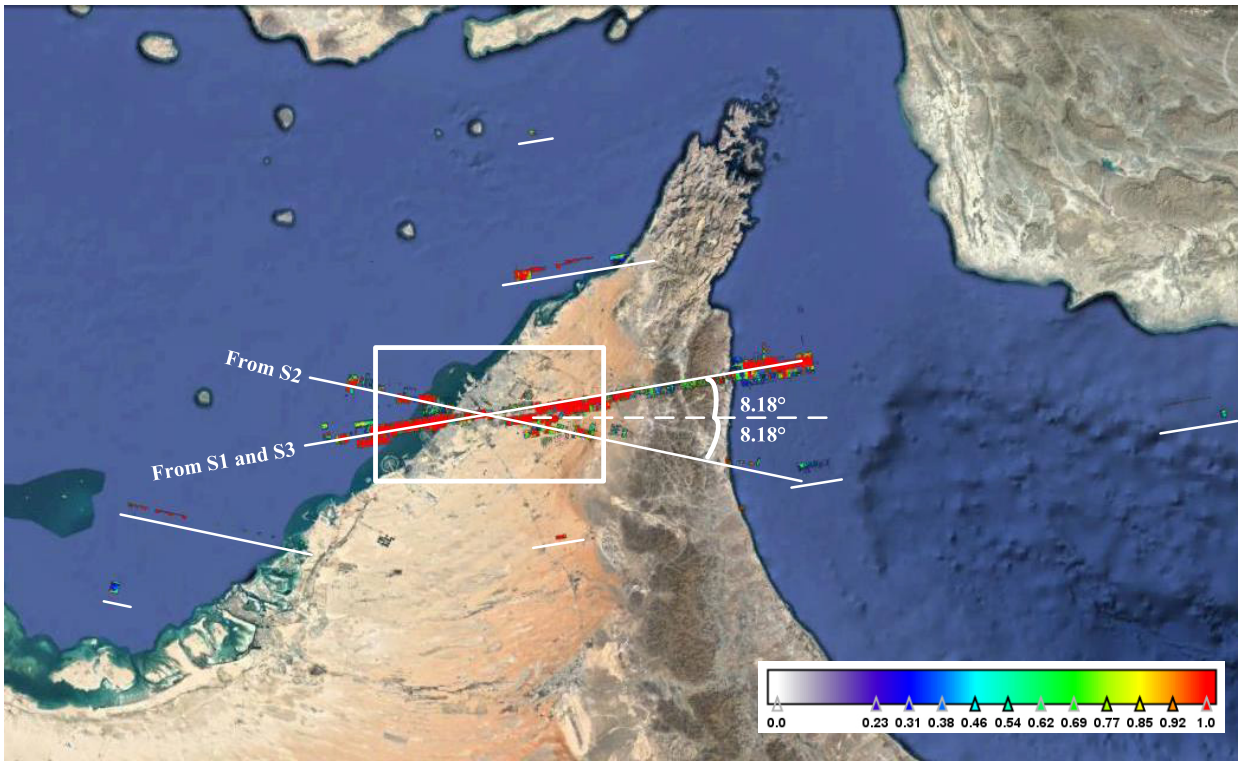


Fig. 13. RFI detection results overlaid on the Google Earth. The white solid lines mark the detected RFIs. The white rectangle indicates the intersection.

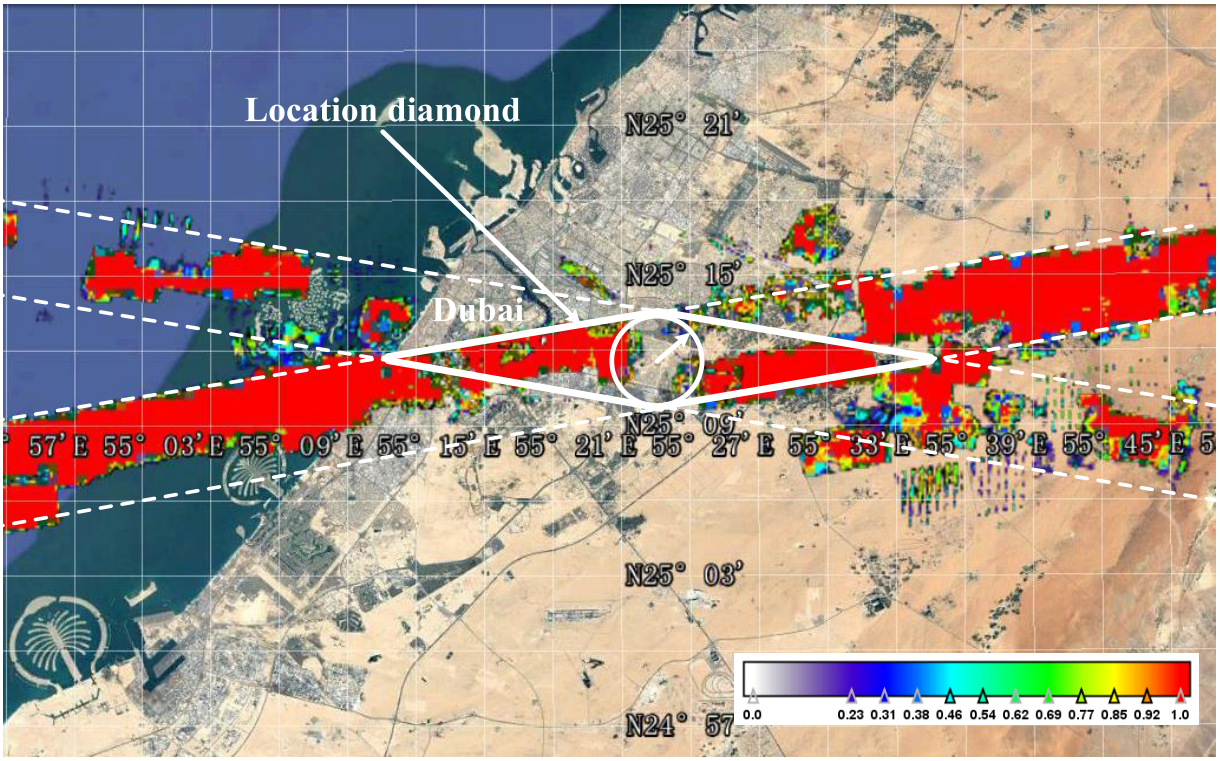


Fig. 14. RFI localization results of the experimental data. The location is marked with the white diamond. The center is approximately at 25°11′ 43.59′ N, 55°25′ 43.87″ E.

degrade the detection performance. Moreover, as our method depends on the assumption that RFI is from the ground source, the localization method is invalid for RFI from other satellites, e.g., the RadarSAT-2 or GF-3.

V. Conclusion

In this article, we presented a detailed characterization of RFI based on the generation mechanism and proposed a new method for RFI detection and localization in Sentinel-1

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.

images. Different from conventional methods, the proposed method works on GRD images. A powerful index for RFI detection, i.e., the RFII, was first proposed. It is designed based on the dual-polarization difference of RFI. An RFI localization method was also provided, which assumes that the presence of RFI is persistent in ascending–descending interval and utilizes the RFI detection results in different passes. The location area is a diamond of approximately 88.76 km^2. Experimental results based on real Sentinel-1 data demonstrated that the proposed method can achieve good performance. It is expected to be able to play an important role in monitoring RFI and qualifying Sentinel-1 data.

ACKNOWLEDGMENT

The authors would like to thank the European Space Agency (ESA) for providing free Sentinel-1 data and software online.

REFERENCES

[1] I. Cumming and F. Wong, *Digital Signal Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Norwood, MA, USA: Artech House, 2004.

[2] J. Lee and E. Pottier, *Polarimetric Radar Imaging: From Basic to Applications*. Boca Raton, FL, USA: CRC Press, 2009.

[3] C. Oliver and S. Quegan, *Understanding Synthetic Aperture Radar Images*. Raleigh, NC, USA: Scitech, 2004.

[4] A. Monti-Guarnieri, D. Giudici, and A. Recchia, "Identification of C-Band radio frequency interferences from Sentinel-1 data," *Remote Sens.*, vol. 9, no. 11, p. 1183, Nov. 2017.

[5] T. Zhao, Y. Zhang, L. Yang, Z. Dong, and D. Liang, "The RFI suppression method based on STFT applied to SAR," *Prog. Electromagn. Res. M*, vol. 31, pp. 171–188, 2013.

[6] Y. Huang, G. Liao, J. Xu, and J. Li, "Narrowband RFI suppression for SAR system via efficient parameter-free decomposition algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3311–3322, Jun. 2018.

[7] L. H. Nguyen, M. D. Dao, and T. D. Tran, "Joint sparse and low-rank model for radio-frequency interference suppression in ultra-wideband radar applications," in *Proc. 48th Asilomar Conf. Signals, Syst. Comput.*, Nov. 2014, pp. 864–868.

[8] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Eigensubspace-based filtering with application in narrow-band interference suppression for SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 75–79, Jan. 2007.

[9] S. Zhang, M. Xing, R. Guo, L. Zhang, and Z. Bao, "Interference suppression algorithm for SAR based on time–frequency transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3765–3779, Oct. 2011.

[10] J. T. Johnson *et al.*, "The CubeSat radiometer radio frequency interference technology validation (CubeRRT) mission," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 299–301.

[11] F. J. Meyer, J. Nicoll, and A. P. Doulgeris, "Characterization and correction of residual RFI signatures in operationally processed ALOS PALSAR imagery," in *Proc. EUSAR, 9th Eur. Conf. Synth. Aperture Radar*, Nuremberg, Germany, Apr. 2012, pp. 83–86.

[12] F. J. Meyer, J. B. Nicoll, and A. P. Doulgeris, "Correction and characterization of radio frequency interference signatures in L-band synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 4961–4972, Oct. 2013.

[13] Y. Soldo *et al.*, "Localization of RFI sources for the SMOS mission: A means for assessing SMOS pointing performances," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 2, pp. 617–627, Feb. 2015.

[14] Y. Soldo, P. de Matthaeis, D. le Vine, and P. Richaume, "Comparison of L-band RFI from SMOS and aquarius observations," in *Proc. 2nd SMOS Sci. Conf., ESA-ESAC*, Villafranca, Madrid, May 2015, pp. 25–29.

[15] S. Chan, M. Fischman, and M. Spencer, "RFI mitigation and detection for the SMAP radar," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 2725–2728.

[16] F. Ticconi, C. Anderson, J. F. Saldana, and J. Wilson, "Analysis of the noise scenario measured by ASCAT," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4882–4885.

[17] F. Ulaby and D. A. Lang, "Committee on a survey of the active sensing-uses of the radio spectrum board on physics and astronomy," in *A Strategy for Active Remote Sensing Amid Increased Demand for Radio Spectrum*. Washington, DC, USA: The National Academy Press, 2015.

[18] P. Potin, B. Rosich, J. Roeder, and P. Bargellini, "Sentinel-1 mission operations concept," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1465–1468.

[19] P. Potin and B. Rosich, "Sentinel-1 Mission Status," in *Proc. EUSAR, 11th Eur. Conf. Synth. Aperture Radar*, Hamburg, Germany, Oct. 2016, pp. 1–6.

[20] (2020). *Sentinels Scientific Data Hub*. [Online]. Available: https://scihub.copernicus.eu/

[21] H. Greidanus and C. Santamaria, "First analyses of Sentinel-1 images for maritime surveillance," in *JRC Science and Policy Reports*. Luxembourg: Publications Office of the European Union, 2014.

[22] C. Santamaria, M. Alvarez, H. Greidanus, V. Syrris, P. Soille, and P. Argentieri, "Mass processing of Sentinel-1 images for maritime surveillance," *Remote Sens.*, vol. 9, no. 7, p. 678, Jul. 2017.

[23] V. S. Krishna Vanama and Y. S. Rao, "Change detection based flood mapping of 2015 flood event of chennai city using Sentinel-1 SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 9729–9732.

[24] J. Singh, A. Mahapatra, S. Basu, and B. Banerjee, "Assessment of Sentinel-1 and Sentinel-2 satellite imagery for crop classification in indian region during kharif and rabi crop cycles," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 3720–3723.

[25] S. Wei, H. Zhang, C. Wang, L. Xu, F. Wu, and B. Zhang, "Large-scale Rice mapping of thailand using Sentinel-1 multi-temporal SAR data," in *Proc. SAR Big Data Era (BIGSARDATA)*, Aug. 2019, pp. 1–4.

[26] H. Zhang, H. Cao, C. Wang, Y. Dong, B. Zhang, and L. Li, "Detection of land use change in urban agglomeration using Sentinel-1 SAR data," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Aug. 2019, pp. 1–4.

[27] P. Meadows, A. Pilgrim, R. Piantanida, D. Riva, and N. Miranda, "Sentinel-1A radiometric calibration," in *Proc. CEOS Workshop Group Calibration Validation, SAR Subgroup, Synth. Aperture Radar Workshop*, Noordwijk, The Netherlands, Oct. 2015, pp. 1–34. [Online]. Available: http://sarcv.ceos.org/site_media/media/documents/CEOS_SAR_Workshop_Sentinel-1A_Meadows.pdf

[28] (2017). *Sentinel Toolbox Exploitation Platform Forum*. [Online]. Available: http://forum.step.esa.int/t/noise-objects-in-sentinel-1-grd-data/7627

[29] X. Leng, K. Ji, S. Zhou, X. Xing, and H. Zou, "Discriminating ship from radio frequency interference based on noncircularity and non-Gaussianity in Sentinel-1 SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 352–363, Jan. 2019.

[30] X. Leng, K. Ji, S. Zhou, and X. Xing, "Ship detection based on complex signal kurtosis in single-channel SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6447–6461, Sep. 2019.

[31] X. Leng, K. Ji, S. Zhou, and X. Xing, "Fast shape parameter estimation of the complex generalized Gaussian distribution in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 11, pp. 1933–1937, Nov. 2020.

[32] J. Stastny *et al.*, "A novel adaptive synthetic aperture radar ship detection system," in *Proc. OCEANS MTS/IEEE KONA*, Sep. 2011, pp. 1–7.

[33] H. Greidanus, M. Alvarez, C. Santamaria, F.-X. Thoorens, N. Kourti, and P. Argentieri, "The SUMO ship detector algorithm for satellite radar images," *Remote Sens.*, vol. 9, no. 3, p. 246, Mar. 2017.

[34] X. Leng, K. Ji, K. Yang, and H. Zou, "A bilateral CFAR algorithm for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1536–1540, Jul. 2015.

[35] X. Leng, K. Ji, X. Xing, S. Zhou, and H. Zou, "Area ratio invariant feature group for ship detection in SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 1–13, Apr. 2018.

[36] X. Yang, V. Carcione, F. Nunziata, M. Portabella, and M. Migliaccio, "Assimilation of SAR-derived sea surface winds into typhoon forecast model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 6060–6062.

[37] T. Miller, L. Potter, and J. McCorkle, "RFI suppression for ultra wideband radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 4, pp. 1142–1156, Oct. 1997.

[38] M. Vreugdenhil *et al.*, "Sensitivity of Sentinel-1 backscatter to vegetation dynamics: An Austrian case study," *Remote Sens.*, vol. 10, no. 9, p. 1396, Sep. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.



Xiangguang Leng (Member, IEEE) received the B.S. degree in remote sensing science and technology from Wuhan University (WHU), Wuhan, China, in 2013, and the M.S. degree (Hons.) in photogrammetry and remote sensing and the Ph.D. degree in electronic science and technology from the National University of Defense Technology (NUDT), Changsha, China, in 2015 and 2019, respectively.

He is a Lecturer with the College of Electronic Science and Technology, NUDT. He has authored more than 30 articles. His research interests mainly include remote sensing information processing, synthetic aperture radar (SAR) image interpretation, and machine learning.

Dr. Leng was a recipient of the Excellent Paper Award of the 2016 CIE International Conference on Radar, the Excellent Paper Award of the 2018 IET International Radar Conference, and the Excellent Paper Award of the 5th China High Resolution Earth Observation Conference. He is a reviewer for several international journals.



Kefeng Ji (Member, IEEE) received the B.S. degree in aerospace engineering from Northwestern Polytechnical University (NWPU), Xi'an, China, in 1997, and the M.S. and Ph.D. degrees in information and telecommunication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 1999 and 2003, respectively.

In 2003, he joined the College of Electronic Science and Technology, NUDT, where he is a Professor. He has authored or coauthored over 80 articles. His research interests include signal processing, machine learning, pattern recognition, remote sensing information processing, synthetic aperture radar (SAR) image interpretation, target detection, recognition, feature extraction, and marine surveillance.

Dr. Ji is a reviewer for several international journals and conferences, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE ACCESS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, *International Journal of Remote Sensing*, *Remote Sensing Letters*, *European Journal of Remote Sensing*, and *Remote Sensing*.



Gangyao Kuang (Senior Member, IEEE) received the B.S. and M.S. degrees in geophysics from the Central South University of Technology, Changsha, China, in 1988 and 1991, respectively, and the Ph.D. degree in communication and information from the National University of Defense Technology (NUDT), Changsha, in 1995.

He is a Professor with the College of Electronic Science and Technology, NUDT. His research interests include remote sensing, synthetic aperture radar (SAR) image processing, change detection, SAR ground moving target indication, and classification with polarimetric SAR images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:15:37 UTC from IEEE Xplore. Restrictions apply.